

Sine rule



- 1 a No b Yes c Yes d No

2 a $\frac{a}{\sin A} = \frac{b}{\sin B}$ b $\frac{p}{\sin P} = \frac{q}{\sin Q} = \frac{r}{\sin R}$

3 a $\sin A = \frac{x}{b}$ b $x = a \sin B$ c $\frac{\sin A}{a} = \frac{\sin B}{b}$

Unknown sides

4 a $a = 10.50$ b $a = 6.85$ c $a = 3.02 \text{ m}$ d $a = 711.73 \text{ cm}$
e $a = 11.29 \text{ cm}$

5 a $x = 13.1$ b $x = 12.9$

6 a i $a = 16.66$ ii $a = 16.66$ b i $a = 17.14$ ii $a = 17.14$
c i $a = 30.09$ ii $a = 30.09$

7 a $HK = 733.17 \text{ km}$ b $KJ = 444.55 \text{ km}$

8 a $\angle A = 75.78$ b $b = 3.152 \text{ m}$ c $c = 5.723 \text{ m}$

Unknown angles

9 a $x = 26.0^\circ$ b $x = 32.7^\circ$ c $x = 50.7$ d $x = 30.4^\circ$
e $x = 20.4^\circ$ f $x = 30.3^\circ$ g $x = 43.2^\circ$ h $x = 149.1^\circ$

i $x = 133.13^\circ$

j $x = 23.5^\circ$



10

a $B = 74^\circ$

b $B = 31^\circ$

11

a $x = 19.54^\circ$

b 48.46°

12

a $x = 41$

b 139

13 $\theta = 33.84^\circ$

Applications

14

a $\angle OBA = 39^\circ$

b $k = 21.76$

15 $x = 53.6$ m

16

a $\angle A = 38^\circ$

b $c = 22.71$ m

c $b = 22.71$ m

d 1.9 min